



DO-IT-YOURSELF: PROJECT

simulated. In each case, the system detects abnormal values and responds appropriately, validating the predictive failure mechanisms.

Software

Control code and web monitoring. This system uses two distinct code modules. The first runs on the Arduino to handle PID-based feedback and motor control. The second is an optional web server code to enable remote monitoring. An ESP8266 module is required for this functionality.

PID Arduino control code. The control code uses the PID library, which can be installed via the Library Manager in the Arduino IDE. The Arduino continuously reads sensor inputs, calculates PID output, and adjusts motor speed based on encoder feedback. Select the correct board and port

to upload the code. The sensors provide real-time data for predictive fault detection. System outputs are visible via the serial monitor for local observation.

Optionally, the ESP8266 module can transmit data to a web server, enabling users to access system status in real time through a browser interface.

Construction and testing

Begin by uploading the source code to the Arduino Uno using the appropriate board and port settings. Assemble the circuit as shown in Fig. 2, then power the system.

After powering on, the rotary encoder monitors rotation and detects missed steps. The system applies the PID formula to correct them and sends correction signals to the motor driver via pin 9 as an

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analogue output. The temperature sensor monitors the motor drive system. If excessive heat is detected, the system issues a warning through the serial monitor and flags a potential failure. **EFY**

Rajesh Krishnan is an embedded systems engineer passionate about cutting-edge technologies. He specialises in IoT and motor control firmware, with ongoing research in innovative embedded solutions.



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